

Tailoring Machine Learning Weather Predictions for Local Impacts

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Research Goal and Overview

- Show the feasibility of finetuning weather forecasts for specific outcomes using minimal compute power
- Demonstrate the method using 2 meter temperature and 10 meter wind speed across a variety of regions and lead times
- 2m Temp Improvements range from XX-XX
- 10m Wind Speed Improvements range from XX-XX
- Show that an ML forecast like Pangu can be corrected in the same way as traditional NWP forecast like IFS

Key Contribution: Demonstrating a simple method to significantly improve forecast accuracy using limited computational power.

Previous that Post-Processing NWP Forecasts with Neural Networks is Effective

- citehanDeepLearningMethod2021 trained a U-Net to correct IFS Forecasts in Northern China using 12 years of data and a NVIDIA GPU
 - Improvements in 2m Temp RMSE go range from 30% to 12% for lead times 1 to 7 days
 - Improvements in 10m Wind Speed RMSE Range from 25% to 20% for lead times 1 to 7 days
- citearaujoPostProcessingAirTemperature2022 test using both a CNN and MLP to further improve post processing model GRIDPP over Norway
- Uses RMSE error instead of MSE but I find a larger percent improvement for both 2m temperature and 10m wind speed

Method

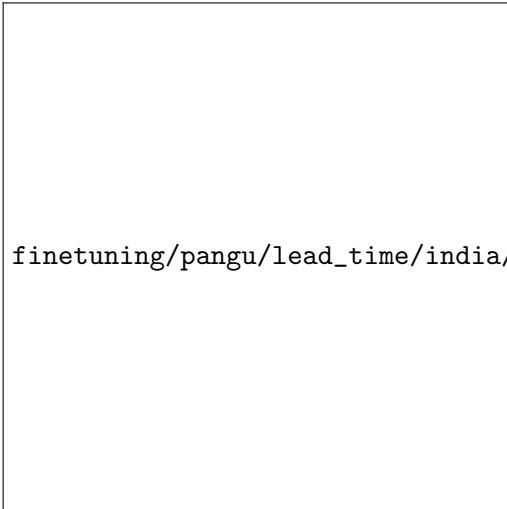
- Uses Pangu and IFS HRES forecasts with lead times from 1 to 7 days
 - Target Pangu predictions using ERA5 and IFS HRES using its $t=0$ initialization
- Filter to a single variable and lead time, in a particular region
- Train simple MLP network that takes in Pangu/IFS forecast outputs and minimizes the forecast error
 - Train using a single forecast per day for years 2018-2021
 - Test using daily forecasts from 2022

Headline Result: Simple MLP Seems Effective at Correcting Forecasts

- Simple MLP method works across variety of regions and climates
- Training a correction model for a 10x10 degree region takes about 2.5 minutes on an M3 max MacBook Pro
- Similar improvements to both Pangu and IFS forecasts

finetuning/pangu/comparison/mse_comparison_pangu_pangu_nempatho2/mse_comparison

See Similar Percent Improvements Across Different Lead Times



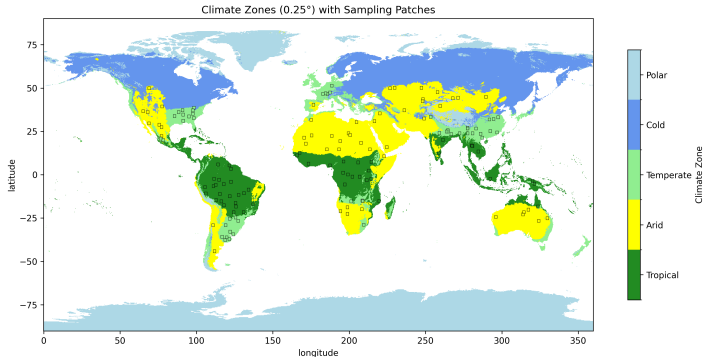
(a) 2m Temperature: 10x10 degree



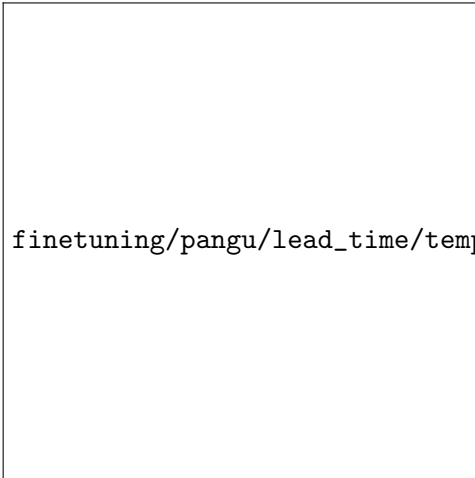
(b) Wind Speed: 10x10 degree

How to Explore Heterogeneity by Climate Region

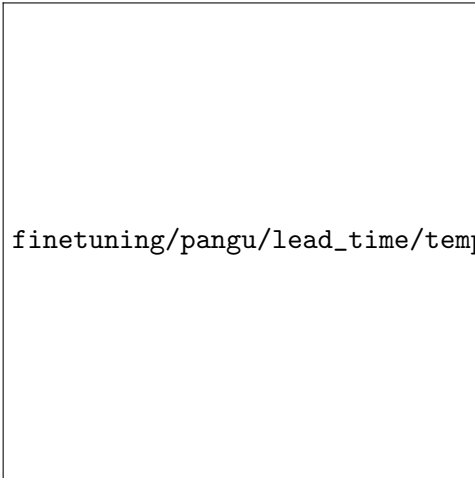
- Define tropical, temperate, and arid pixels on an 0.25 degree grid using broad Koppen-Geiger classes
- Train fine-tuning model on randomly sampled 2x2 degree patches



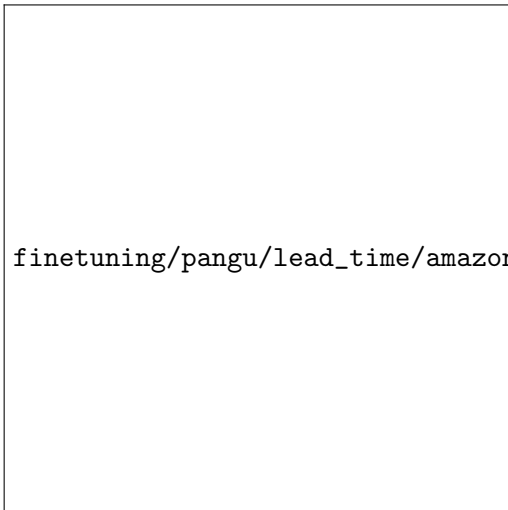
Heterogeneity by Climate Region: 2m Temperature



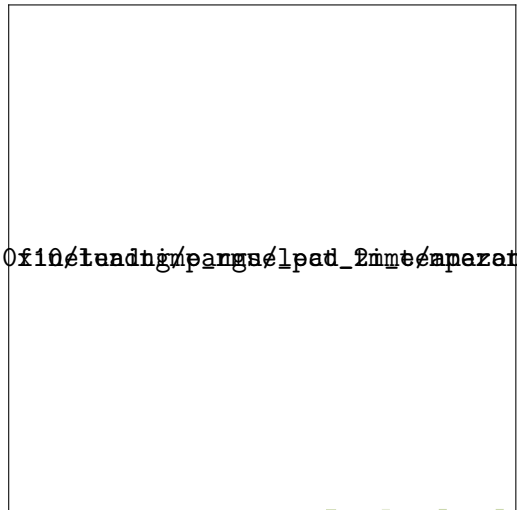
Heterogeneity by Climate Region: 10m Wind Speed



Amazon



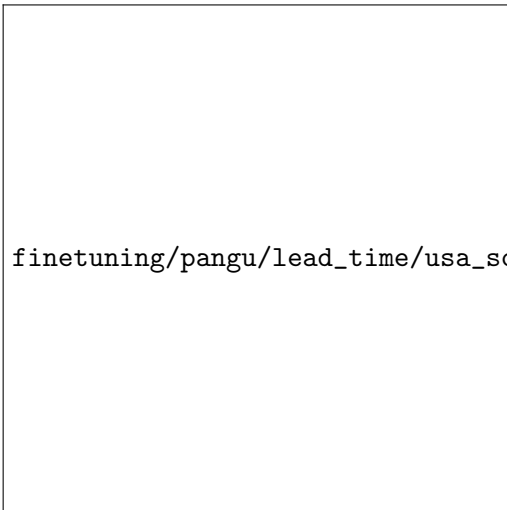
(a) 2m Temperature: 10x10 degree



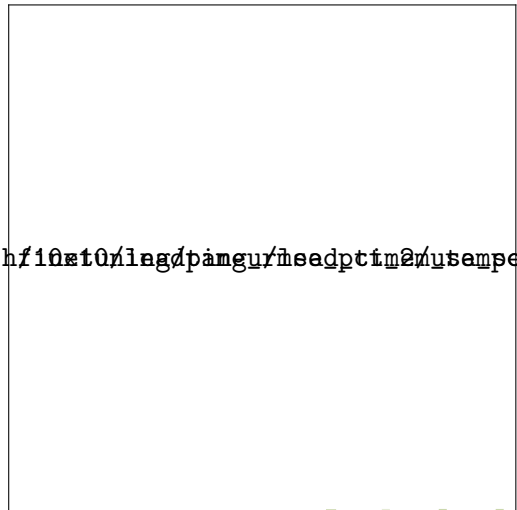
(b) Wind Speed: 10x10 degree

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US South

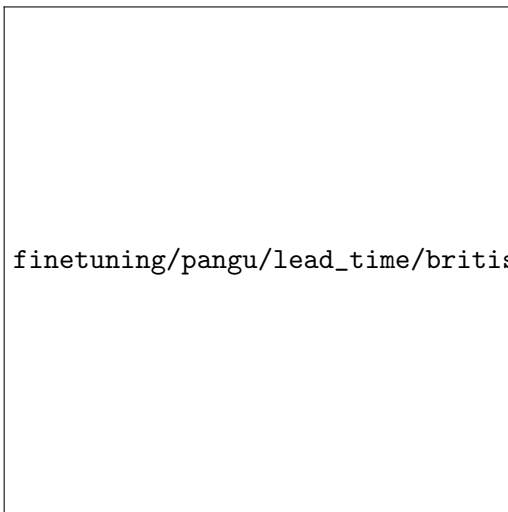


(a) 2m Temperature: 10x10 degree

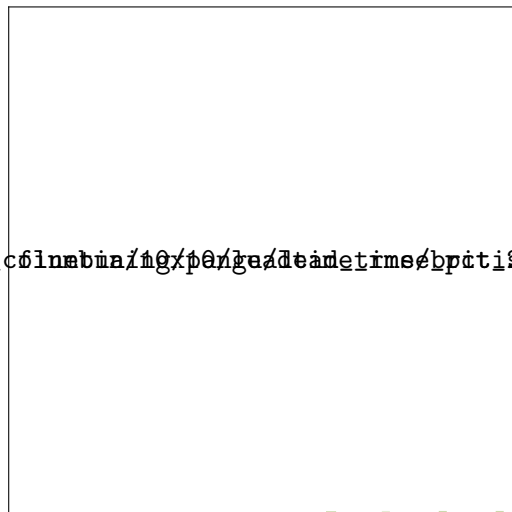


(b) Wind Speed: 10x10 degree

British Columbia



(a) 2m Temperature: 10x10 degree



(b) Wind Speed: 10x10 degree

What I'm Actively Working on

- Rewriting the databuild to be able to take advantage of the new WeatherbenchX package and also be able to run on a server and be more flexible
- Implementing a U-Net model for fine-tuning
 - Early results suggest it significantly ups compute time without much improved performance
- Implement ANO Correction (subtracting mean error) method to compare to MLP correction